WINTER SEMESTER 2026

Foundations of Biomedical Informatics

Lecture 7: SNOMED CT

Table 1. Types of Terminologies Used for Computational Analysis.*

Term	Definition
Index	List of relevant terms pulled directly from a body of unstructured or semistructured text. An index is produced to improve the speed and relevance of search results.
Terminology	Set of preferred or official terms in a domain. A terminology may be a systematic nomenclature supported by a centralized body or as simple as the common usage that arises in a specific community of practice.
Thesaurus	Terminology that clusters synonyms and plesionyms (near synonyms) into categories.
Controlled vocabulary	Set of preferred terms created specifically for a domain or body of text.
Classification	Controlled vocabulary that is intended to comprehensively describe a topic or domain from a conceptual perspective and is not developed solely from a text corpus that it is meant to describe.
Statistical classification	Classification in which all concepts are mutually exclusive to avoid counting anything twice. This is typically achieved with the use of a monohierarchy, in which each concept has one and only one parent, such as the *International Classification of Diseases* (ICD).[7] A statistical classification is exhaustive because it includes residual categories such as "unspecified" or "not elsewhere classified."
Ontology	Controlled terminology invoking formal semantic relationships between and among concepts, manifested as a type of description logic, which is a subset of first-order predicate logic, chosen to accommodate computational tractability.[8] A common example is OWL (Web Ontology Language; www.w3.org/OWL/).

Table 2. Standards, Terminologies, and Ontologies Widely Used in Clinical Medicine.*

Type of Data	Ontology	Example of Term
Diagnoses	Systematized Nomenclature of Medicine Clinical Terms (SNOMED CT) ICD Orphanet Rare Disease Ontology (ORDO) National Cancer Institute Thesaurus (NCIT)	Triple-negative breast carcinoma (NCIT:C71732)
Phenotypic abnormalities	Human Phenotype Ontology (HPO)	Bronchopulmonary sequestration (HP:0010960)
Medications	RxNorm DrugBank ChEMBL	Panobinostat (CHEMBL483254)
Adverse reactions	Ontology of Adverse Events (OAE)	Injection-site induration (OAE:0000323)
Procedures	Medical Dictionary for Regulatory Activities (MedDRA)	Cardiac aneurysm repair (MEDDRA/10007514)
Laboratory examinations	Logical Observation Identifiers Names and Codes (LOINC)	Creatinine in serum or plasma (LOINC:2160-0)
Imaging data	Digital Imaging and Communications in Medicine (DICOM) RadLex	Periosteal cortical thinning (RID45761)

Interoperating with Human Entered Data (Text)

```
> dat[1,2]
[1] "A 23-year-old white female presents with complaint of allergies."
> dat[1,5]
[1] "SUBJECTIVE:,  This 23-year-old white female presents with complaint of allergies.  She used to have allergies when s
he lived in Seattle but she thinks they are worse here.  In the past, she has tried Claritin, and Zyrtec.  Both worked fo
r short time but then seemed to lose effectiveness.  She has used Allegra also.  She used that last summer and she began 
using it again two weeks ago.  It does not appear to be working very well.  She has used over-the-counter sprays but no p
rescription nasal sprays.  She does have asthma but doest not require daily medication for this and does not think it is 
flaring up.,MEDICATIONS: , Her only medication currently is Ortho Tri-Cyclen and the Allegra.,ALLERGIES: , She has no kno
wn medicine allergies.,OBJECTIVE:,Vitals:  Weight was 130 pounds and blood pressure 124/78.,HEENT:  Her throat was mildly
erythematous without exudate.  Nasal mucosa was erythematous and swollen.  Only clear drainage was seen.  TMs were clear.
,Neck:  Supple without adenopathy.,Lungs:  Clear.,ASSESSMENT:,  Allergic rhinitis.,PLAN:,1.  She will try Zyrtec instead 
of Allegra again.  Another option will be to use loratadine.  She does not think she has prescription coverage so that mi
ght be cheaper.,2.  Samples of Nasonex two sprays in each nostril given for three weeks.  A prescription was written as w
ell."
```

SNOMED CT

- Most comprehensive multilingual clinical healthcare terminology and ontology
- > 80 countries use it in EHR systems for reporting and documentation
- Translated into 11 languages
- Concepts are related to each other, grouped and analysed
- Simple application: A Coding System for machine reading clinical documents
- Sophisticated applications: Compositional grammar with complex expressions

Why SNOMED CT Matters in the Real World

- Problem:
- Doctors write free text; computers need structure.

- SNOMED CT enables:
- • Standardized clinical meaning
- • Decision support
- • Public health analytics
- • AI-ready data

- Key idea:
- SNOMED CT converts human language into computable knowledge.

Example from India: Public Health & ABDM

- Use Case: Disease Surveillance

- Example:
- Typhoid fever / Enteric fever → Single SNOMED CT concept

- Impact:
  - National disease aggregation
  - Outbreak detection
  - ICD-10 reporting

Example from India: Clinical Decision Support

- Use Case: Primary Care & Telemedicine

- Scenario:
- Chest pain + left arm radiation + sweating

- SNOMED CT enables:
  - Post-coordinated expressions
  - Automated risk inference

- Outcome: Early referral, life-saving alerts

Interoperability Across EHRs

- Use Case: Multi-hospital patient care

- Problem:
- MI, Heart attack, Myocardial infarction

- SNOMED CT solution:
- Single concept ID preserves meaning

- Outcome:
- Safer transitions, fewer errors

Origin

- 1955- American College of Pathologists started SNOP (Systematized Nomenclature of Pathology)
- 1965- SNOP published, describes pathology findings using four axes:
  - Topography (anatomic site affected)
  - Morphology (structural changes associated with disease)
  - Etiology (the cause of disease) including organisms
  - Function (physiologic alterations associated with disease)
- SNOP was the first multi-axial coding system used in healthcare

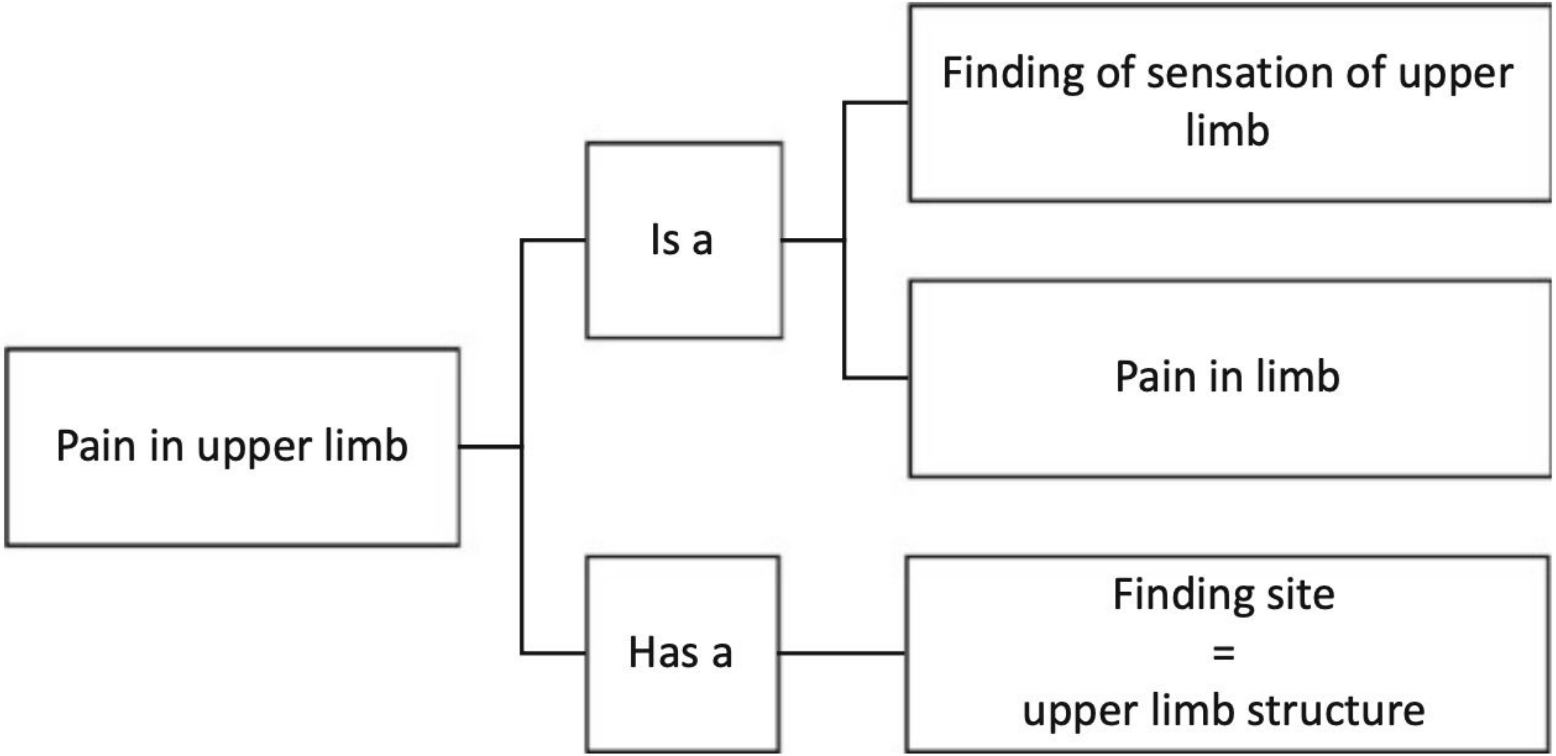
- urine and glucose are *substances*
- urine glucose concentration is an *observable entity*
- a urine glucose test is a *procedure*
- a urine glucose test result is a *clinical finding*
- if a urine glucose test is not done it is a *situation with explicit context*.

SNOMED 3.5 Axes

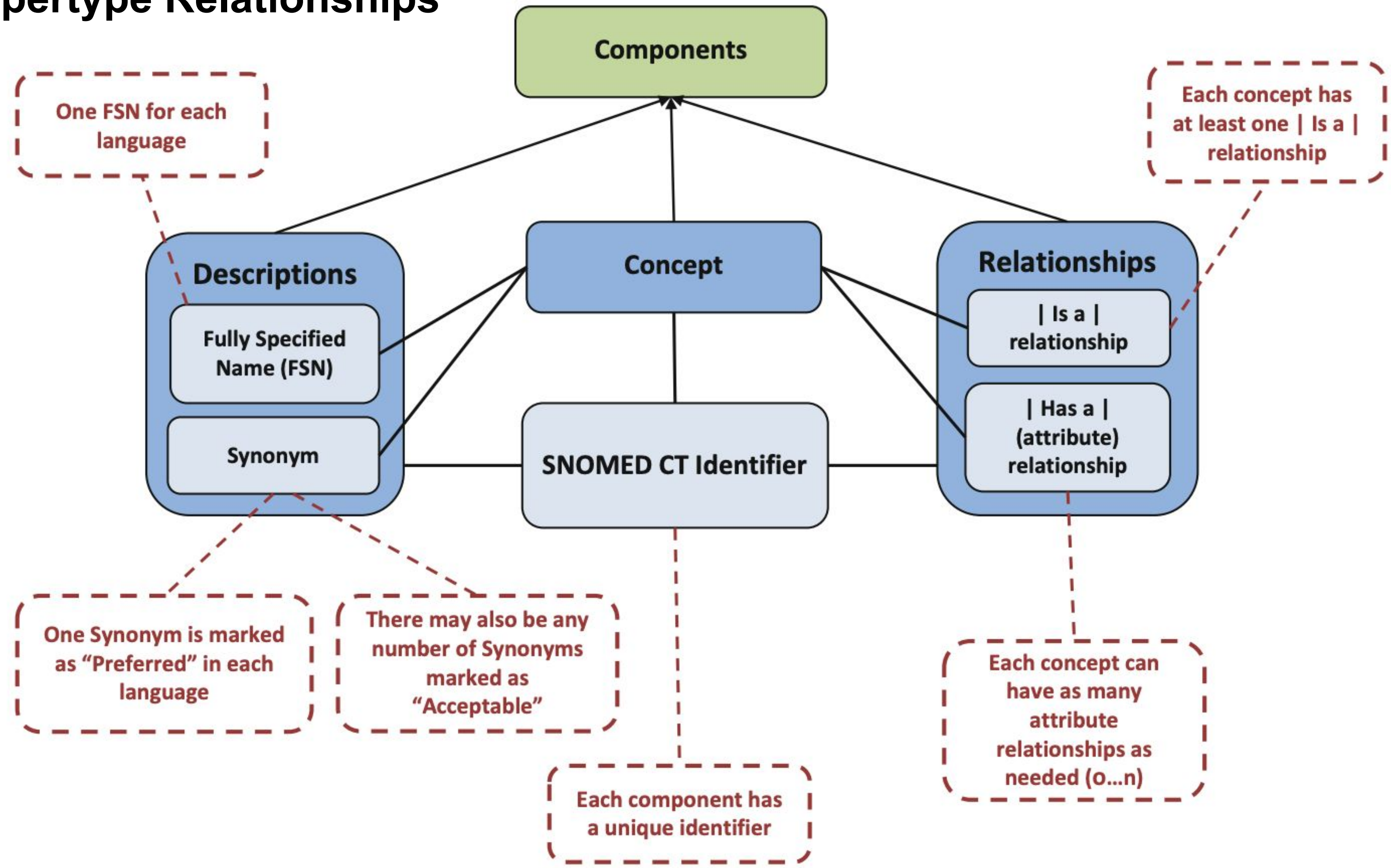
Axis	Description
T	Topography—Anatomic terms (13,000 records)
M	Morphology—Changes found in cells, tissues and organs (6000 records)
L	Living organisms—Bacteria and viruses (25,000 records)
C	Chemical—Drugs (15,000 records)
F	Function—Signs and symptoms (19,000 records)
J	Jobs Terms that describe the occupation (1900 records)
D	Diagnosis—Diagnostic terms (42,000 records)
P	Procedure—Administrative, diagnostic and therapeutic procedures (31,000 records)
A	Agents—Devices, physical agents and activities associated with disease (1600 records)
S	Social context—Social conditions and important relationships in medicine (500 records)
G	General—Syntactic linkages and qualifiers (1800 records)

Concepts, Descriptions

- **Concepts** are clinical meanings that do not change.
- SNOMED CT is concept-oriented
- Concept IDs are machine readable
- Each concept is associated text **descriptions**,
- Descriptions are human readable form of the concept
- Every concept has at least **two descriptions**
  - Fully Specified Name (FSN) - unique and unambiguous, not meant for end users
  - Display term in the language being used
- FSN has a suffix in () indicating its primary hierarchy e.g. myocardial infarction **(disorder)**.
- Display term is often FSN without suffix, e.g. myocardial infarction
- All other descriptions are synonyms
- Synonyms are marked as **Preferred** or **Acceptable**


Pain in upper limb
Is a
Finding of sensation of upper limb
Pain in limb
Has a
Finding site
=
upper limb structure

Supertype Relationships

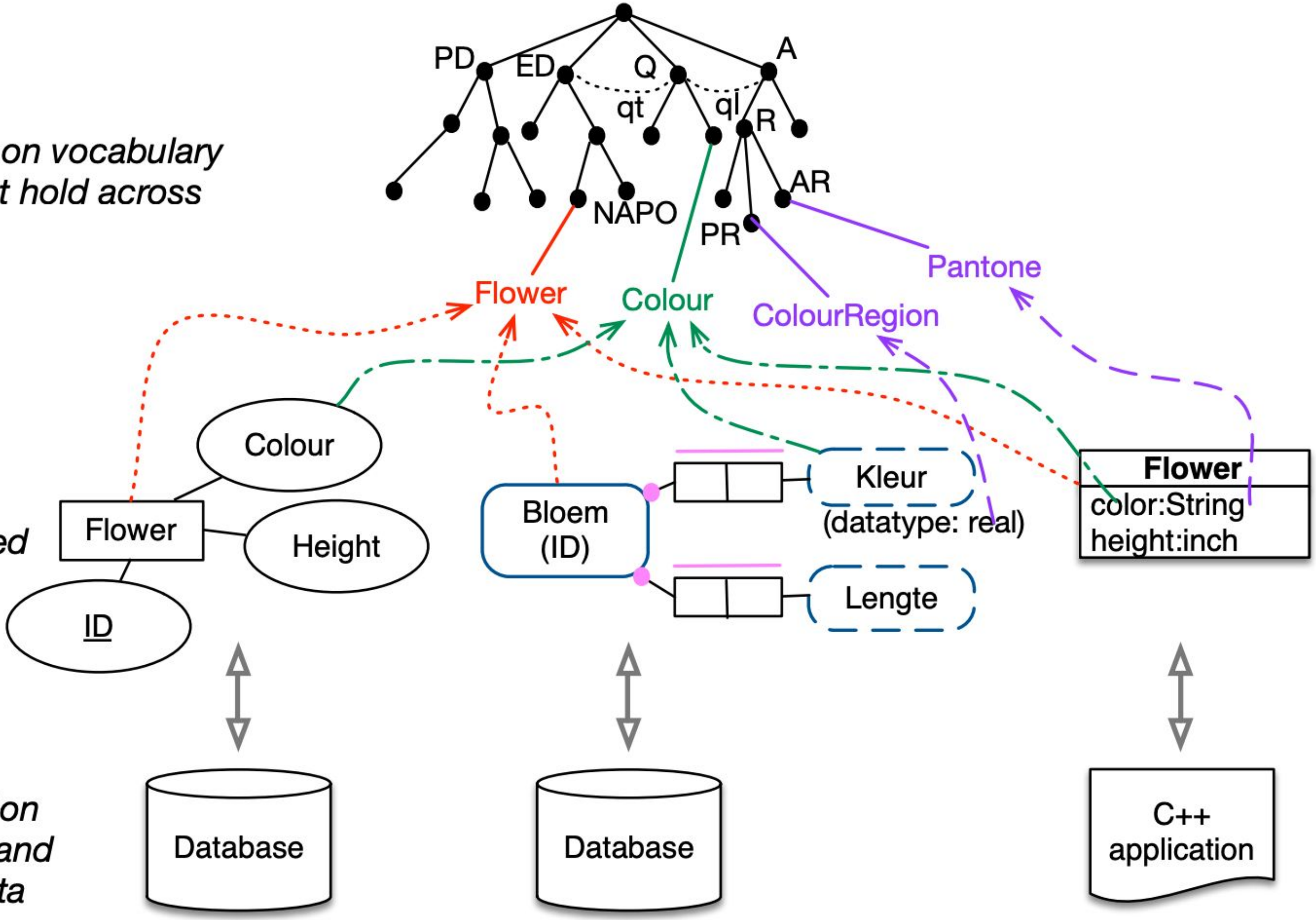


What are Ontologies?

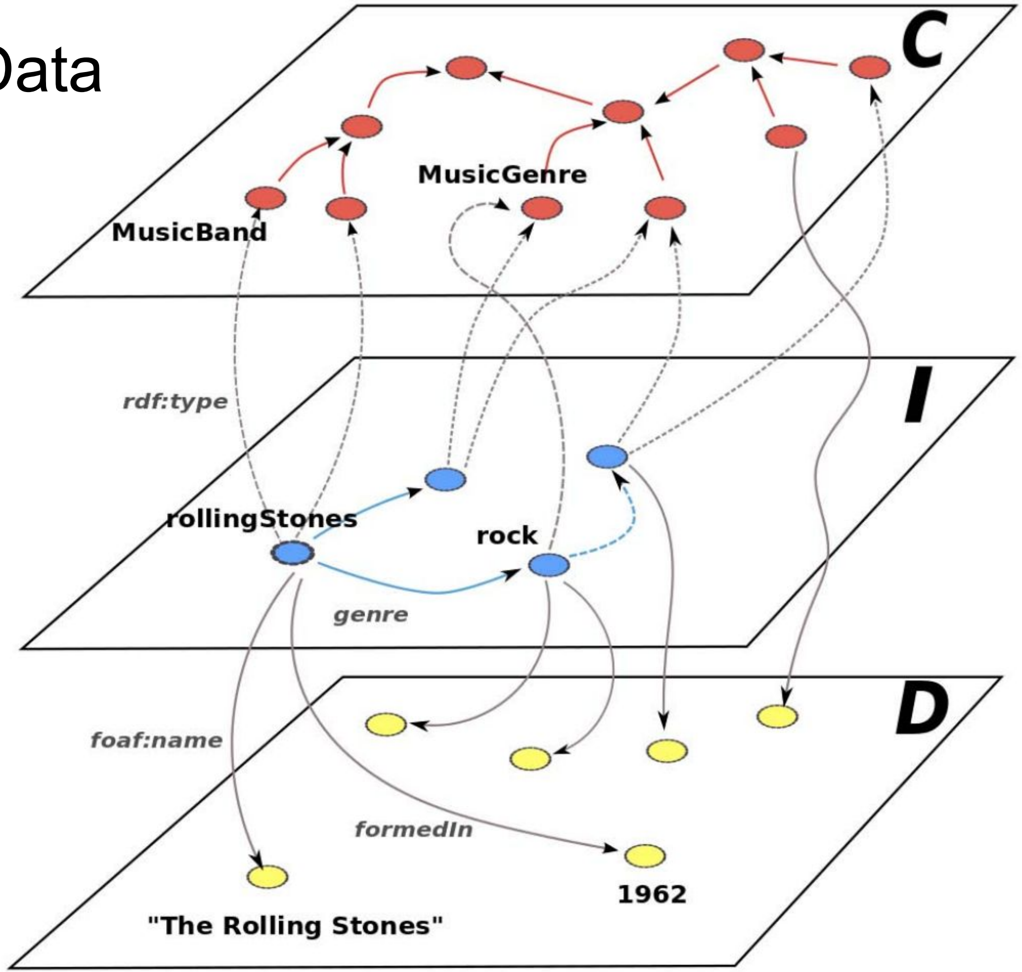
Ontology
provides the common vocabulary and constraints that hold across the applications

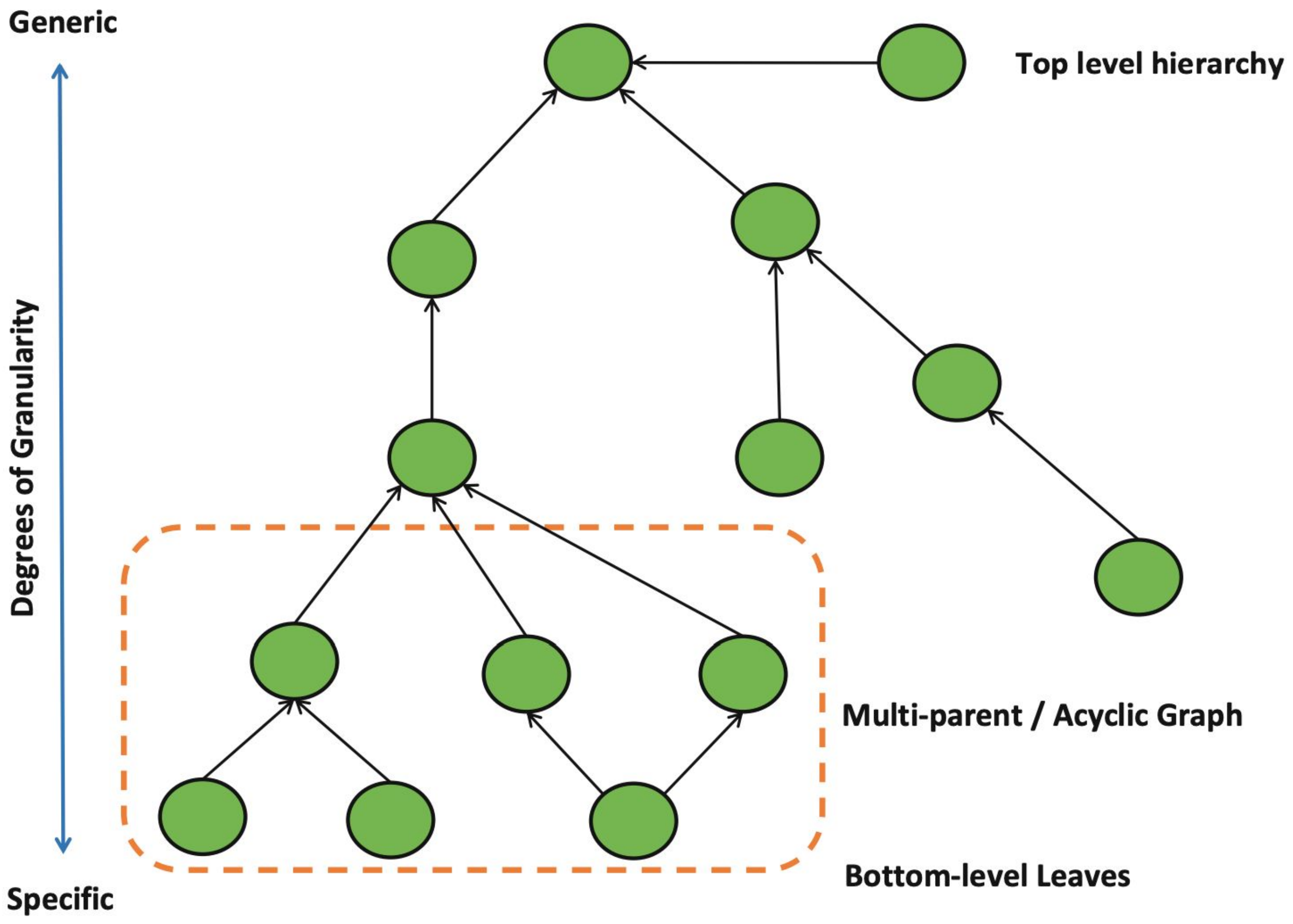
Conceptual model
shows what is stored in that particular application

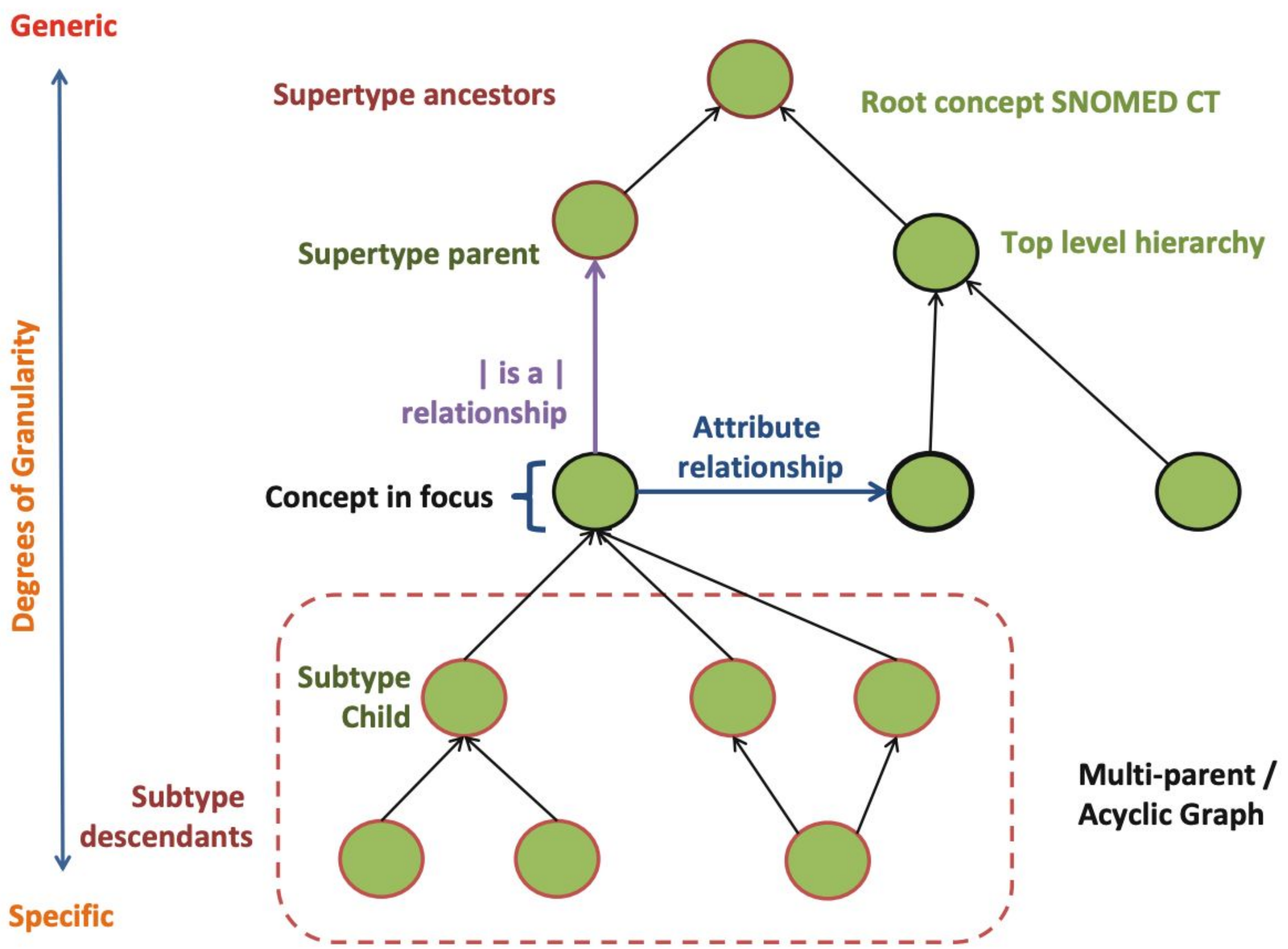
Implementation
the actual information system that stores and manipulates the data



Concepts Instances Data




Generic
Top level hierarchy
Degrees of Granularity
Multi-parent / Acyclic Graph
Bottom-level Leaves
Specific


Generic
Supertype ancestors
Root concept SNOMED CT
Supertype parent
Top level hierarchy
| is a |
relationship
Attribute
relationship
Concept in focus
Degrees of Granularity
Subtype
Child
Subtype
descendants
Multi-parent /
Acyclic Graph
Specific

Defining Relationships

- (1) Supertype Relationships and (2) Defining Attributes
- Everything is a concept
- E.g. |is a| Relationship -> Concept ID: 116680003
- Every concept except the SNOMED CT root has a supertype
- A concept is **sufficiently defined** if its defining relationships are sufficient to distinguish it from all its supertype and sibling concepts.
- Another concept represented as a combination of the same defining characteristics, is equivalent to it or a subtype of it.
- Large parts of SNOMED CT are not yet sufficiently defined.
- Primitive concepts: not fully defined and do not have the unique relationships needed to distinguish them from their parent or sibling concepts.
- E.g. Pneumonia - unless defining characteristics are specified

Expressions

- Pre-coordinated: when a single concept identifier is used to represent a clinical idea
- Post-coordinated: to represent a meaning using a combination of two or more concept identifiers
- ID |Name|
- Refinement : (colon)
    - Attribute (Can be more than one, separated by comma)
    - Value

```
80146002|appendectomy|:260870009|priority|=25876001|emergency|
```

```
80146002|appendectomy|:260870009|priority|=25876001|emergency|,
425391005|using access device|=86174004|laparoscope|
```

Description Logic

|concept|:|attribute|=|value|

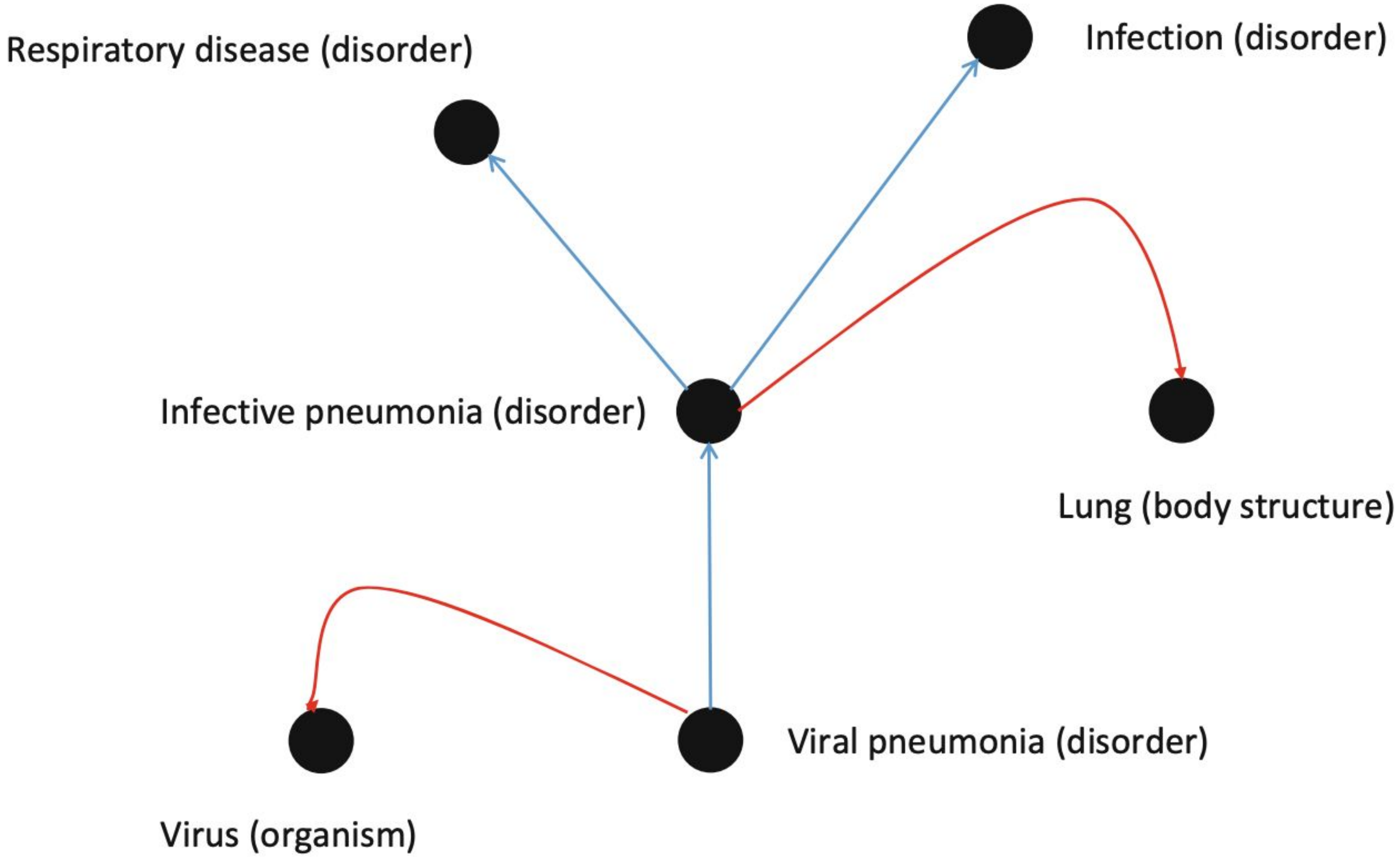
|Viral pneumonia (disorder)| |is a| |Infective pneumonia (disorder)|

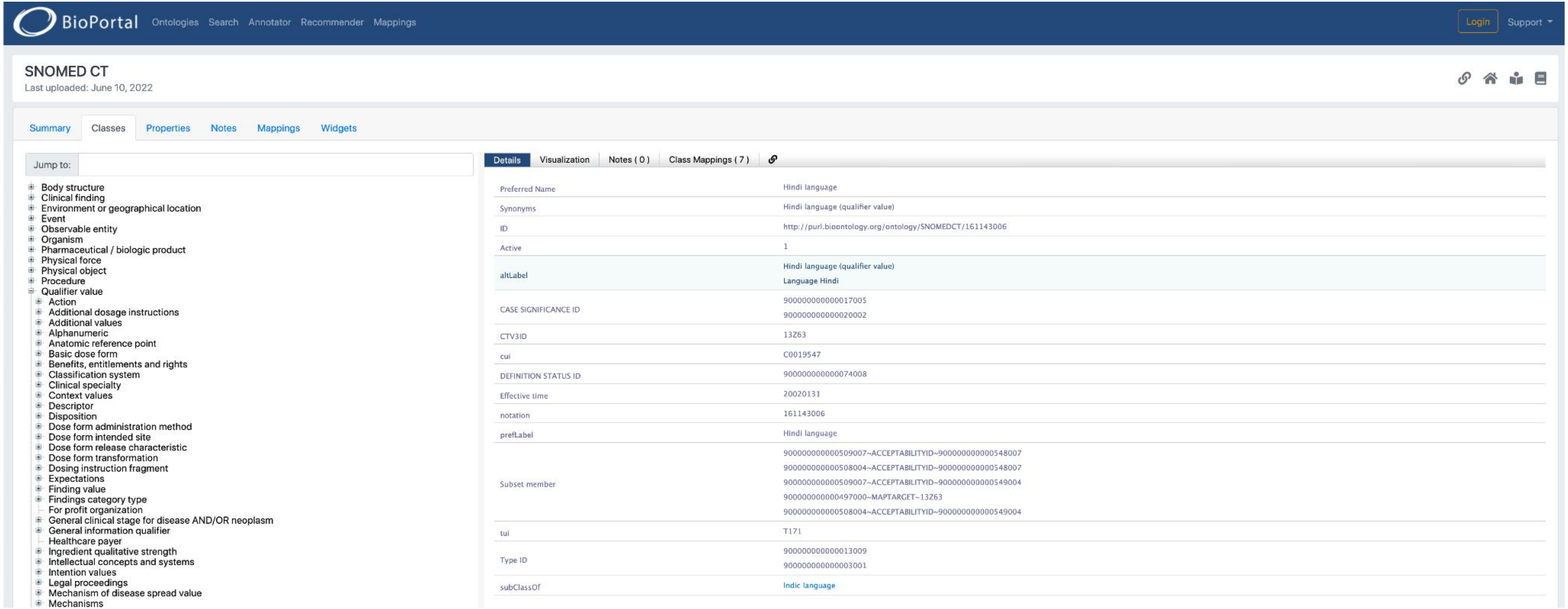
|Viral pneumonia (disorder)| "has a" |Causative agent (attribute)| = |Virus (organism)|

|Infective pneumonia (disorder)| |is a| |Respiratory disease (disorder)|

|Infective pneumonia (disorder)| |is a| |Infection (disorder)|

|Infective pneumonia (disorder)| "has a" |Finding site (attribute)| = |Lung (body structure)|


Respiratory disease (disorder)
Infection (disorder)
Infective pneumonia (disorder)
Lung (body structure)
Viral pneumonia (disorder)
Virus (organism)



https://bioportal.bioontology.org/ontologies/SNOMEDCT?p=classes&conceptid=161143006

snomedizer-package: snomedizer: R Interface to the SNOMED-CT Terminology Server...

In ramses-antibiotics/snomedizer: R Interface to the SNOMED-CT Terminology Server REST API

Description | SNOMED CT licensing | Licensing for reference use | Author(s) | See Also

Description

Manipulate the SNOMED CT clinical ontology using the SNOMED International Terminology Server REST API <https://github.com/IHTSDO/snowstorm>. logo

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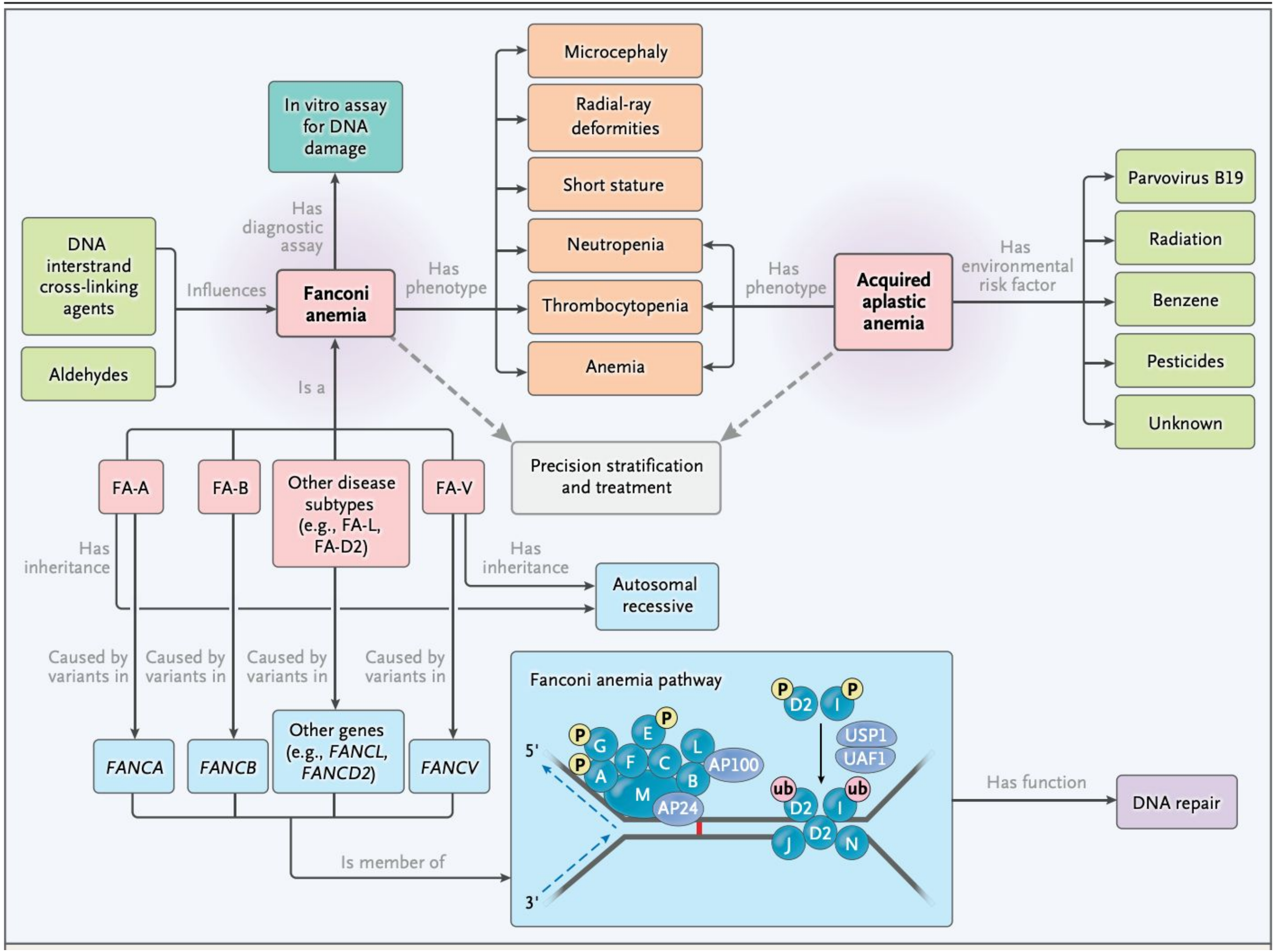
Automated Reasoning



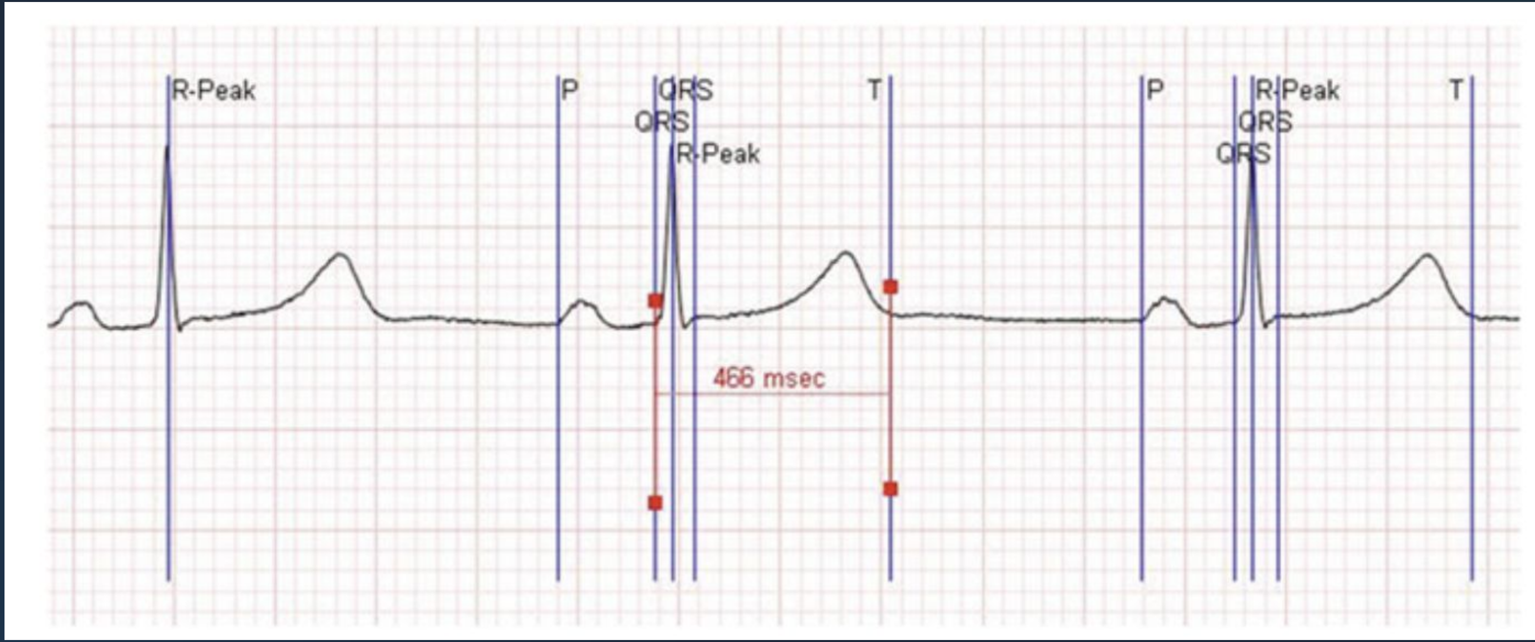
SNOMED CT India Reference Sets

- Developed and released by NRCeS (India).

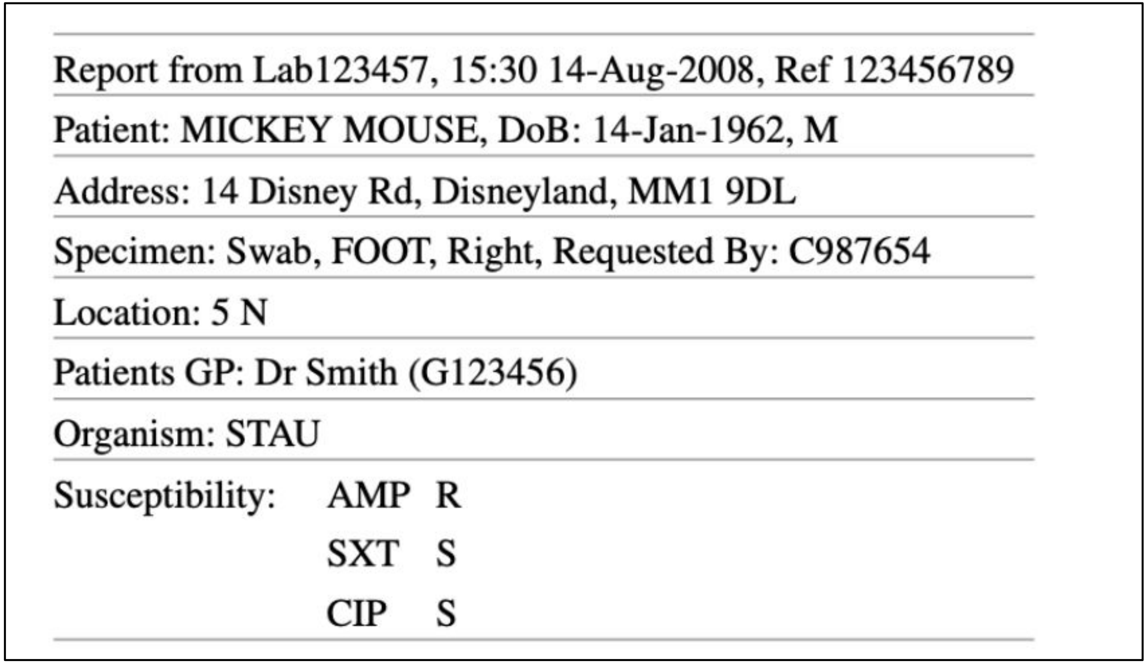
- Clinical domains include:
  - Tuberculosis, Dengue, Malaria
  - Stroke, Cervical cancer
  - Child health & infectious diseases

- These are production-grade clinical terminology artifacts.

Interoperating with Machine Data



What we want



Report from Lab123457, 15:30 14-Aug-2008, Ref 123456789

Patient: MICKEY MOUSE, DoB: 14-Jan-1962, M

Address: 14 Disney Rd, Disneyland, MM1 9DL

Specimen: Swab, FOOT, Right, Requested By: C987654

Location: 5 N

Patients GP: Dr Smith (G123456)

Organism: STAU

Susceptibility:	AMP	R
	SXT	S
	CIP	S

What it looks like at the backend

```
MSH|^~\&||^123457^Labs|||200808141530||ORU^R01|12345678
  9|P|2.4
PID|||123456^^^SMH^PI||MOUSE^MICKEY||19620114|M|||14
  Disney Rd^Disneyland^^^MM1 9DL
PV1|||5 N|||||G123456^DR SMITH
OBR|||54321|666777^CULTURE^LN|||20080802||||||||SW^^^FOO
  T^RT|C987654
OBX||CE|0^ORG|01|STAU||||||F
OBX||CE|500152^AMP|01||||R|||F
OBX||CE|500155^SXT|01||||S|||F
OBX||CE|500162^CIP|01||||S|||F
```

Technical Interoperability: The Technology Layer

This is the technology layer

- Moves data from system A to system B, neutralizing the effects of distance
- Domain independent
- Does not know or care about the meaning of what is exchanged
- Information theory is the foundation stone of technical interoperability

Health Level 7 International (HL7)

- Is a Standards Development Organization
- Is now the lingua franca to pass health messages across systems
- At least two translations required during information interchange

Internal format ---> Wire Format ---> Internal Format
(computer) (connection) (computer)

Requires the following sequential steps in Object Oriented Software Development

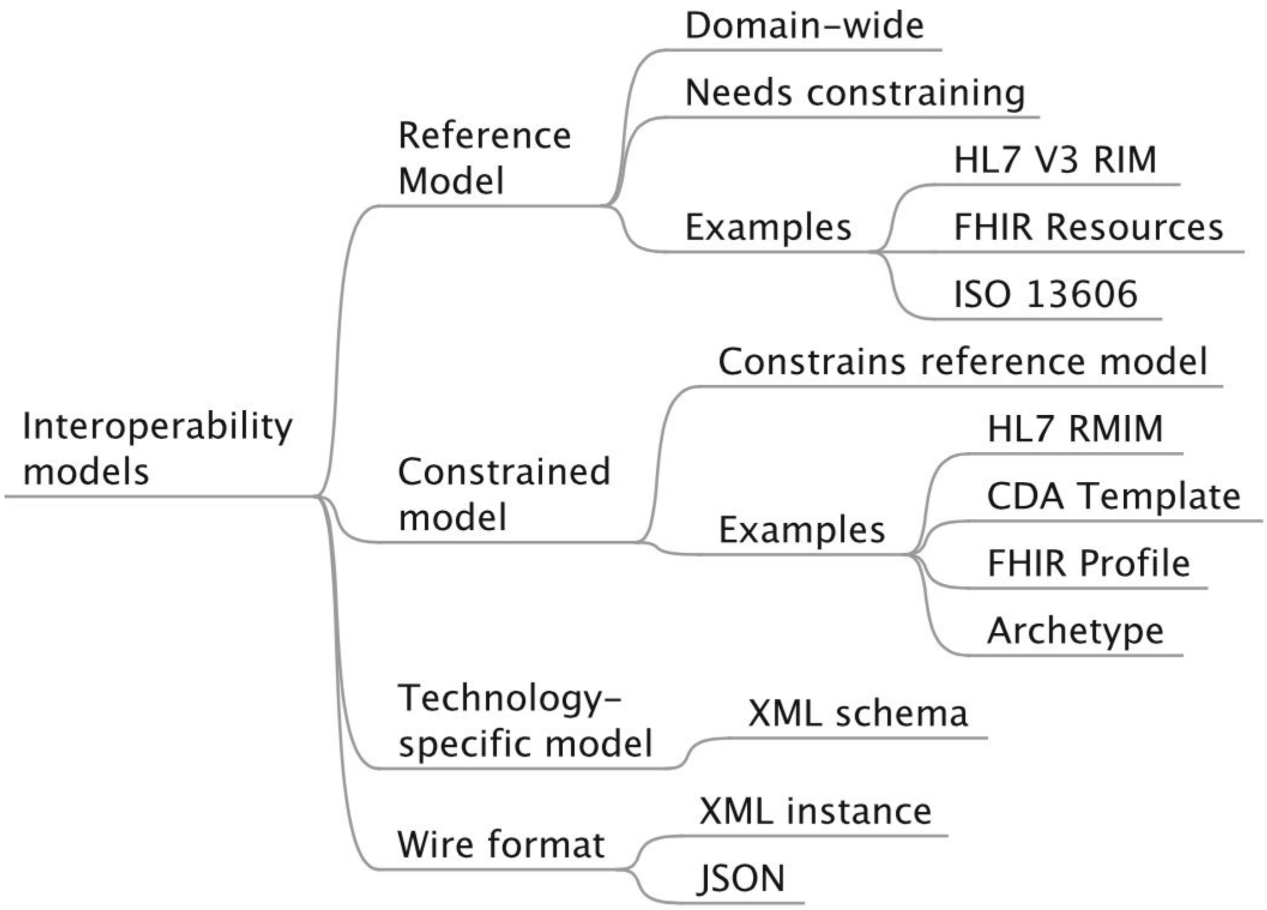
i. Computational-Independent Model (CIM).
ii. Platform-Independent Model (PIM) - the conceptual design of a system.
iii. Platform-Specific Model (PSM)- the implementable design.
iv. Code, the actual software code, also referred to as wire-format.

Another Example of Syntactic Interoperability: HL7 Standard

```
##### Translate this function from HL7 V2.8 ADT_A01 to HL7 V2.4 ADT_A05
### HL7 V2.8 ADT_A01
```

```
    MSH|^~\&|ADT1|GOOD HEALTH HOSPITAL|GHH LAB, INC.|GOOD HEALTH
HOSPITAL|198808181126|SECURITY|ADT^A01^ADT_A01|MSG00001|P|2.8||
    EVN|A01|200708181123||
    PID|1||PATID1234^5^M11^ADT1^MR^GOOD HEALTH
HOSPITAL~123456789^^^USSSA^SS||EVERYMAN^ADAM^A^III||19610615|M||C|222
2 HOME STREET^^GREENSBORO^NC^27401-1020|GL|(555) 555-2004|(555)555-
2004||S||PATID12345001^2^M10^ADT1^AN^A|444333333|987654^NC|
    NK1|1|NUCLEAR^NELDA^W|SPO^SPOUSE||||NK^NEXT OF KIN
    PV1|1|I|2000^2012^01||||004777^ATTEND^AARON^A|||SUR||||ADM|A0|

### HL7 V2.4 ADT_A05
```


Interoperability models
Reference Model
Domain-wide
Needs constraining
Examples
HL7 V3 RIM
FHIR Resources
ISO 13606
Constrained model
Constrains reference model
Examples
HL7 RMIM
CDA Template
FHIR Profile
Archetype
Technology-specific model
XML schema
Wire format
XML instance
JSON

Segment I
Segment

```
MSH|^~\&||^123457^Labs|||200808141530||ORU^R01|12345678
  9|P|2.4
PID|||123456^^^SMH^PI||MOUSE^MICKEY||19620114|M|||14
  Disney Rd^Disneyland^^^MM1 9DL
PV1|||5 N|||||G123456^DR SMITH
OBR|||54321|666777^CULTURE^LN|||20080802||||||||SW^^^FOO
  T^RT|C987654
OBX||CE|0^ORG|01|STAU||||||F
OBX||CE|500152^AMP|01||||R|||F
OBX||CE|500155^SXT|01||||S|||F
OBX||CE|500162^CIP|01||||S|||F
```

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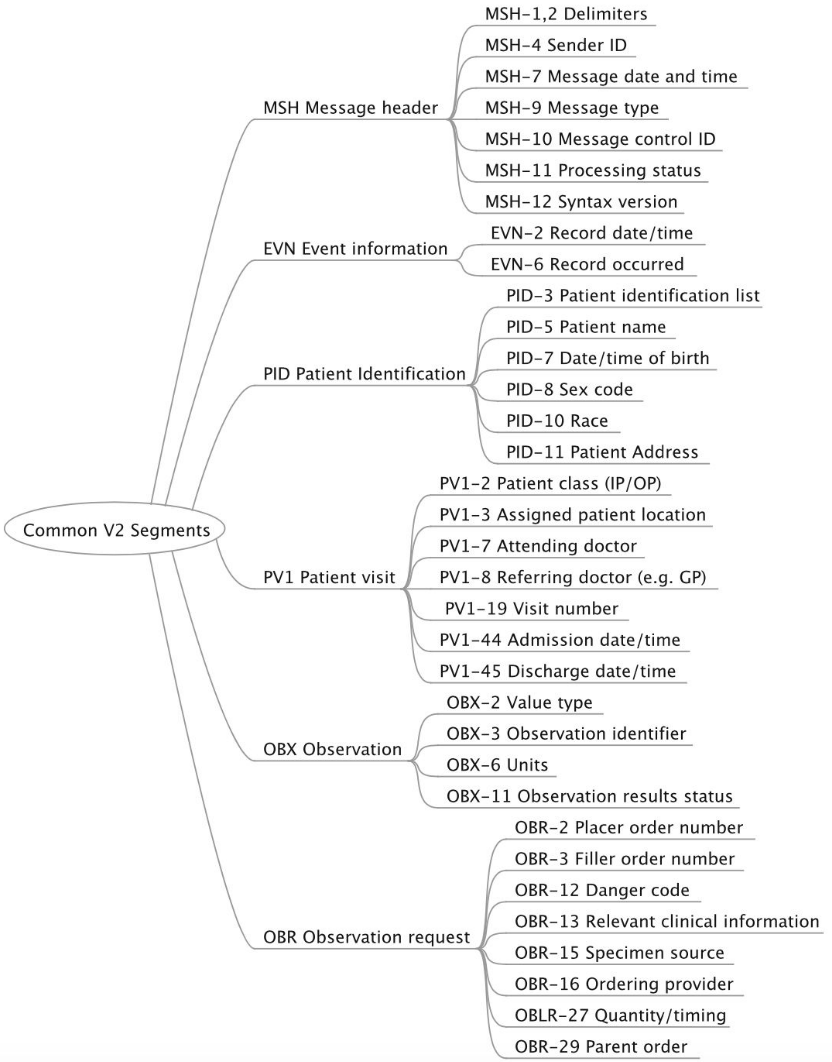
Organism: STAU

Susceptibility:	AMP	R
	SXT	S
	CIP	S



HL7 Mes

Value	Description
ACK	General acknowledgment message
ADT	ADT message
ORM	Order message
ORU	Observation result unsolicited

Value	Description
A01	Admit/visit notification
A02	Transfer a patient
A03	Discharge/end visit
A04	Register a patient

MSH	Message Header
EVN	Event Type
PID	Patient Identification
PV1	Patient Visit

Symbol	Usage
|	Field separator
^	Component separator
~	Repetition separator
\	Escape character
&	Subcomponent separator
<CR>	Segment terminator